# Interim Project Report: Credit Risk Proxy Model Foundation

TO: Bati Bank Executive Team & Project Stakeholders

FROM: Analytics Engineering Team

DATE: December 2025 (Current)

SUBJECT: Completion of Business Understanding, EDA, and Credit Risk Proxy Variable Creation

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## Executive Summary

This report confirms the successful completion of the foundational phases (Tasks 1 & 2) for developing a **Buy-Now-Pay-Later (BNPL) credit risk scoring model**. Given the lack of traditional loan default data, a robust **Credit Risk Proxy Variable** was successfully engineered. This was achieved by analyzing customer transaction behavior using **RFM (Recency, Frequency, Monetary) analysis** and **K-Means Clustering**. The project is now moving into the final phase of model training and validation.

## Section 1: Business Understanding and Regulatory Context

### 1.1 The Necessity of a Proxy Variable

The primary challenge is the absence of a direct historical 'Default' target variable. Therefore, we substituted it with a **Proxy Variable** defined as **Inferred Risk** based on transaction behavior.

- **Proxy Definition:** Customers clustered as having **High Recency** (low activity), **Low Frequency**, and **Low Monetary** value are classified as **High Risk (Proxy = 1)**, reflecting behavior that suggests low commitment or financial constraint.

### 1.2 Model Trade-Offs and Basel II Compliance

In alignment with regulatory best practices, particularly **Basel II** requirements for model auditability and stability, the choice of model balances predictive performance against interpretability.

| Model Type | Advantage (Pro) | Disadvantage (Con) |
|---|---|---|

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|---|---|---|
| **Simple (e.g., Logistic Regression)** | High Interpretability: Satisfies Basel II and is easy to audit and explain to regulators. | Lower Predictive Performance: May be less accurate than complex alternatives. |
| **Complex (e.g., Gradient Boosting)** | Higher Performance: Superior prediction accuracy, minimizing overall loss rates. | Low Interpretability: Operates as a "black box," posing a challenge for regulatory approval. |

**Conclusion:** The initial model will be a **Simple, Interpretable Model** (Logistic Regression) to ensure compliance and transparency.

## Section 2: Data Analysis and Feature Engineering

### 2.1 Key Findings from Exploratory Data Analysis (EDA)

1. **Data Quality:** The dataset is clean, with **zero missing values** across all 95,662 transactions, simplifying the data preparation pipeline.
2. **Customer Behavior:** Over **94%** of transactions are focused on **'financial_services'** and **'airtime'**. This confirms the platform is used for stable, high-frequency, essential purchases, providing a strong basis for behavioral scoring.
3. **Monetary Skewness:** The transaction `Value` is **highly right-skewed** (Mean $\gg$ Median), indicating large outliers. This requires a **logarithmic transformation** during the modeling phase to prevent outliers from distorting the risk assessment.

### 2.2 Creation of the Credit Risk Proxy Variable

The `Credit_Risk_Proxy` target variable was successfully engineered on a customer-level DataFrame via the following steps:

1. **RFM Feature Generation:** Calculated **Recency, Frequency, and Monetary** features for all customers.
2. **Feature Standardization:** Features were scaled to ensure that the large `Monetary` values did not unfairly dominate the clustering process.
3. **K-Means Clustering:** The algorithm segmented customers based on similar RFM behavior.
4. **Final Proxy Mapping:** The cluster identified as the least engaged (High Recency, Low Frequency, Low Monetary) was assigned the binary label **1 (High Risk)**, creating the final target variable for the model.

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## Conclusion and Next Steps

The foundation of the credit scoring model is now robustly established. We have successfully transitioned from raw transaction data to a clean, customer-level dataset containing the necessary behavioral features and a well-defined target variable.

### 3.1 Interim Submission Status

| Deliverable | Status | Notes |
|---|---|---|
| **Task 1: Business Understanding (README)** | **Complete** | Conceptual framework, Basel II compliance strategy, and proxy variable necessity are defined. |
| **Task 2: EDA & Feature Engineering (EDA Notebook)** | **Complete** | Data cleaning, RFM feature creation, and Credit Risk Proxy Variable were successfully generated via K-Means clustering. The final dataset, `customer_rfm_with_proxy.csv`, is saved and ready for modeling. |

### 3.2 Path to Model Development

The project now transitions to the model-building phase. We will implement the baseline model identified in our Task 1 analysis: **Logistic Regression**. This choice prioritizes high **interpretability**—essential for regulatory approval and auditability under Basel II—while providing a strong, simple baseline for credit risk probability estimation.

The immediate next steps will be to:

1. **Split Data:** Separate the RFM features and Proxy target into Training and Testing sets.
2. **Model Training:** Fit the Logistic Regression model to the training data.
3. **Model Evaluation:** Use the testing data to measure the model's performance using metrics like **Accuracy, AUC-ROC, and Precision/Recall**, which will define the model's ability to discriminate between High-Risk and Low-Risk customers.

This successful completion of Tasks 1 and 2 de-risks the remainder of the project by ensuring that the most complex steps—data transformation and target variable creation—are finalized.

Now that the reporting is complete and your data is split (`X_train`, `y_train`), we can proceed to the first action item in the "Next Steps": **Model Training**.